

Mapping acoustic signal features to neural activation Patterns During Music Listening Using fMRI

MEDICAL IMAGE ANALYSIS PROJECT

Shriya - EC23I1008
Shreepal - EC23B1107
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Dataset - OpenNeuro fMRI Music Genre Dataset

<https://openneuro.org/datasets/ds003720/versions/1.0.0>

Dataset Details -

- Participants listened to clips from multiple music genres (e.g., classical, jazz, rock, electronic)
- Whole-brain BOLD fMRI recordings captured during listening.

Data includes -

- fMRI scans
- Stimulus timing information
- Audio files used in the experiment
- Preprocessing metadata

Dataset - OpenNeuro fMRI Music Genre Dataset

Why We Use This Dataset -

- Enables analysis of brain activation patterns during music listening.
- Allows mapping between:
 Acoustic features of music and
 Neural responses in different brain regions

Application in Our Project -

- Extract audio features from music stimuli (tempo, spectral features, rhythm).
- Compare with fMRI activation patterns using GLM analysis.
- Identify which audio characteristics activate specific brain regions.

Research paper

We have identified a similar journal publication, which we will use for reference, in order to implement our project.

<https://share.google/XDpmi7bM50ZmZa6zK>

Correspondence of categorical and feature-based representations of music in the human brain

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Abstract

Introduction: Humans tend to categorize auditory stimuli into discrete classes, such as animal species, language, musical instrument, and music genre. Of these, music genre is a frequently used dimension of human music preference and is determined based on the categorization of complex auditory stimuli. Neuroimaging studies have reported that the superior temporal gyrus (STG) is involved in response to general music-related features. However, there is considerable uncertainty over how discrete music categories are represented in the brain and which acoustic features are more suited for explaining such representations.

Methods: We used a total of 540 music clips to examine comprehensive cortical representations and the functional organization of music genre categories. For this purpose, we applied a voxel-wise modeling approach to music-evoked brain activity measured using functional magnetic resonance imaging. In addition, we introduced a novel technique for feature-brain similarity analysis and assessed how discrete music categories are represented based on the cortical response pattern to acoustic features.

Results: Our findings indicated distinct cortical organizations for different music genres in the bilateral STG, and they revealed representational relationships between different music genres. On comparing different acoustic feature models, we found that these representations of music genres could be explained largely by a biologically plausible spectro-temporal modulation-transfer function model.

Conclusion: Our findings have elucidated the quantitative representation of music genres in the human cortex, indicating the possibility of modeling this categorization of complex auditory stimuli based on brain activity.

KEYWORDS

fMRI, MTF model, music genre, STG

Patient consent: Informed consent was obtained from all participants prior to their participation.

Permission to reproduce material from other sources: Not applicable.

Clinical trial registration: Not applicable.

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Dataset & Tool used -

DATASET -

OpenNeuro Music Genre fMRI Dataset

TOOL USED -

fMRIPrep (Nipype-based standardized preprocessing pipeline)

We used fMRIPrep for standardized preprocessing. The anatomical pipeline successfully generated brain masks, segmented tissues and normalized the brain to MNI space.

The preprocessing pipeline ensures spatial consistency and noise reduction, forming a reliable foundation for voxel-wise modeling

1) ANATOMICAL PREPROCESSING -

1) Skull stripping -

Removes non-brain tissues and keeps only brain

2) Bias field correction -

Fixes uneven brightness in MRI to make tissue intensities consistent

3) Tissue segmentation -

- Divides brain into:
 - Gray Matter (neurons)
 - White Matter (connections)
 - CSF (fluid)

4) Spatial normalization -

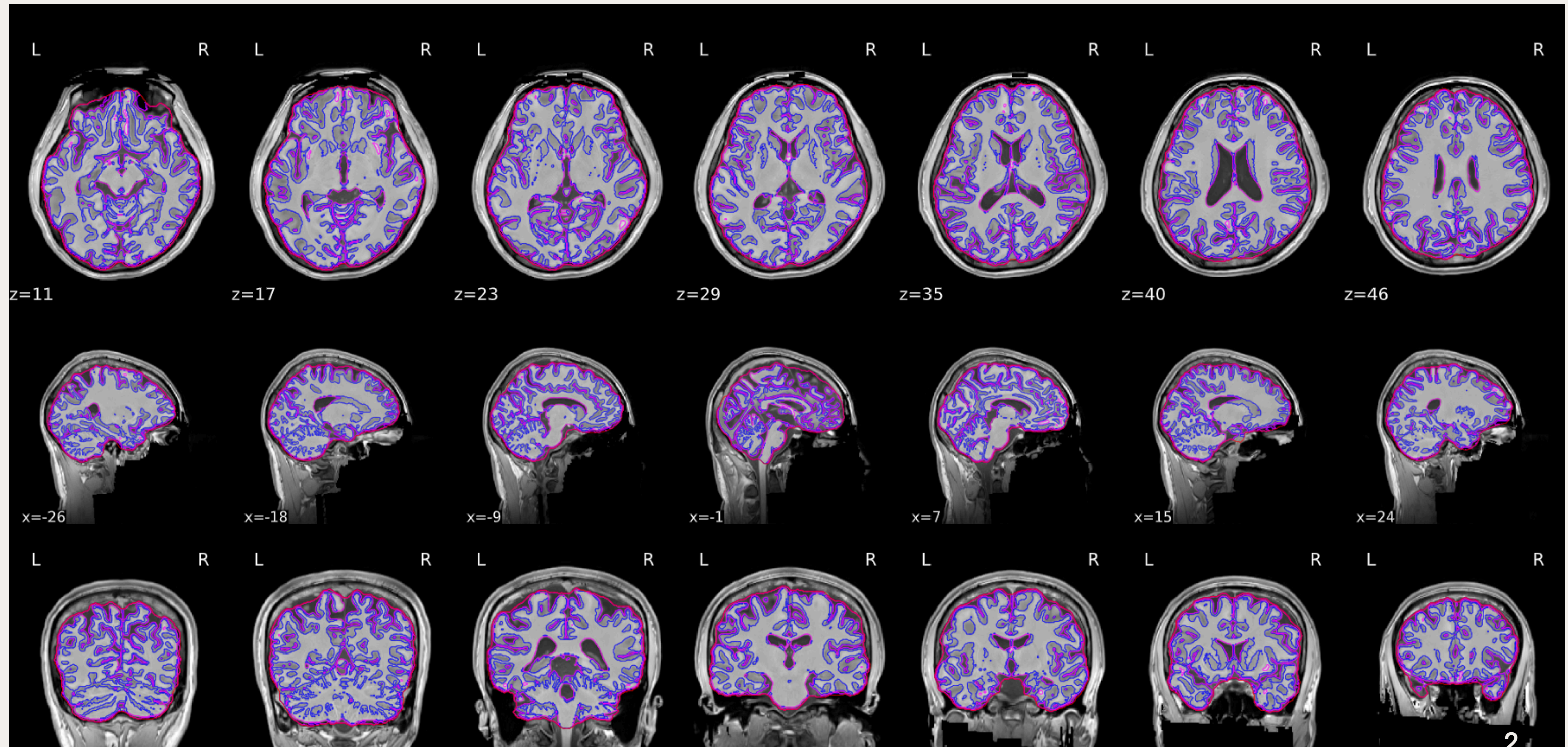
Maps brain → MNI standard space (a standard reference brain)

so that Subject A voxel (x,y,z) = Subject B voxel (x,y,z)

enabling comparison across subjects and mapping to standard brain regions

Brain mask and brain tissue segmentation of the T1w

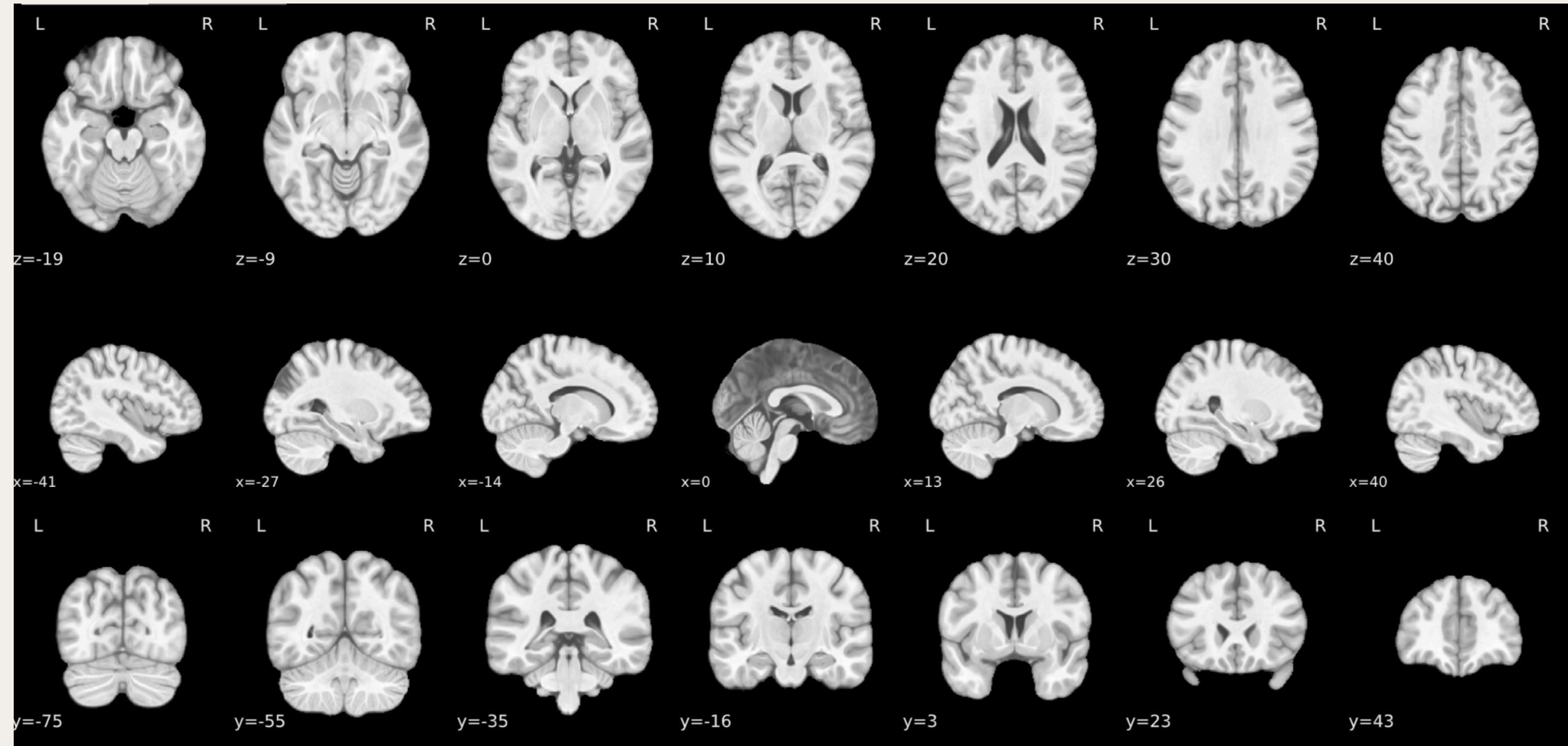
This panel shows the final, preprocessed T1-weighted image, with contours delineating the detected brain mask and brain tissue segmentations.



Observations –

- Gray image → T1 anatomical MRI
- Colored outlines:
 - Purple → Gray Matter (GM)
 - Blue → White Matter (WM)
 - outer boundary → Brain mask
- Multiple slices:
 - axial (top row)
 - sagittal (middle)
 - coronal (bottom)

Spatial normalization of the anatomical T1w reference



2) FUNCTIONAL PREPROCESSING -

Functional preprocessing prepares time-series brain data by correcting temporal misalignment, motion artifacts, and aligning it with anatomical and standard space

1) Slice Timing Correction -

Aligns all slices in a volume to the same acquisition time.

2) Motion Correction

Corrects head movement by aligning all timepoints to a reference volume.

3) Coregistration (BOLD → T1)

Aligns functional data with anatomical images for accurate localization.

3) Spatial Normalization (to MNI)

Maps functional data to a standard brain space for comparison across subjects.

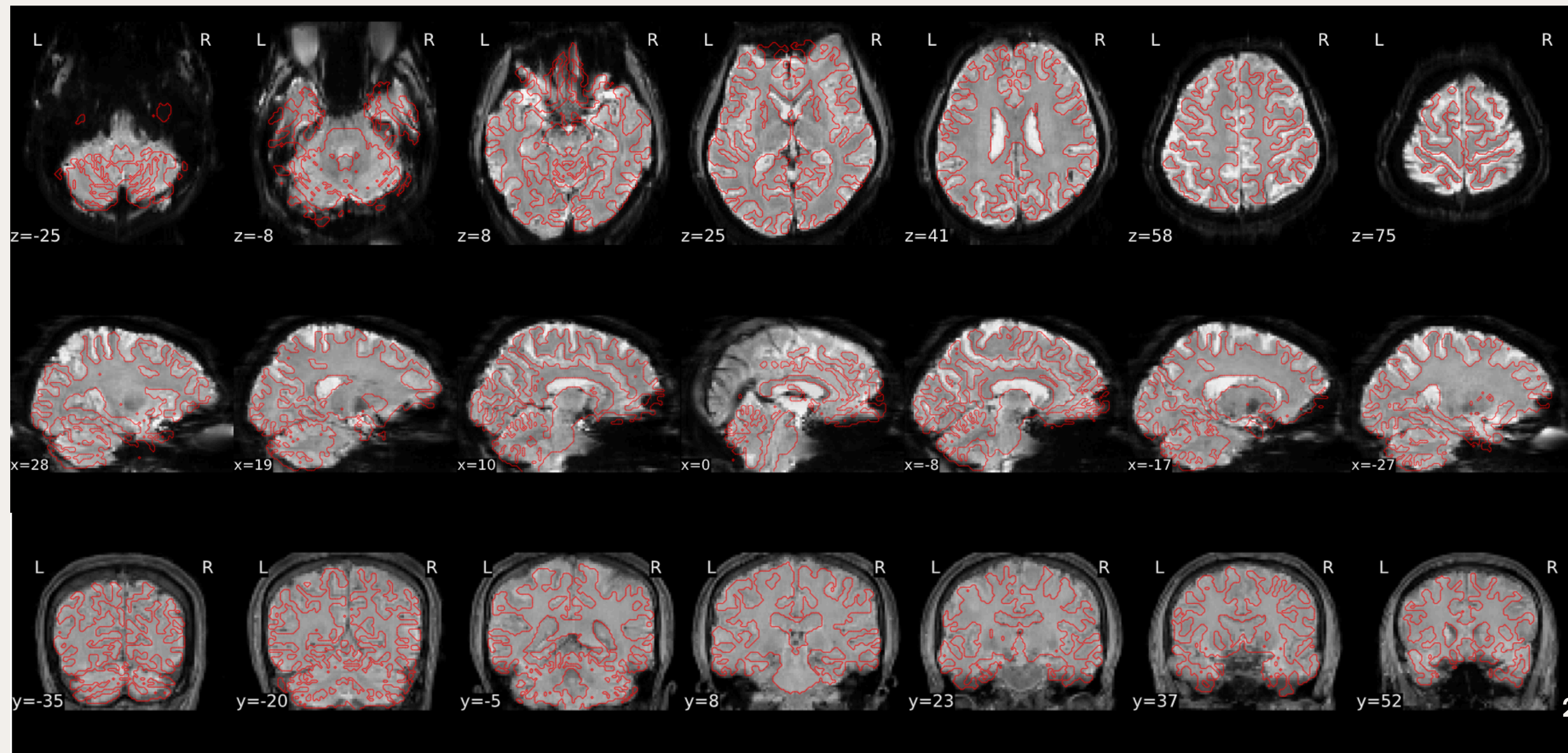
4) Confound Estimation

Identifies and models noise sources like motion and physiological signals.

COREGISTRATION - Alignment of functional and anatomical MRI data

Functional (BOLD) data is aligned to anatomical (T1) image.

Red contours (white matter) show how well the alignment matches the brain structure.



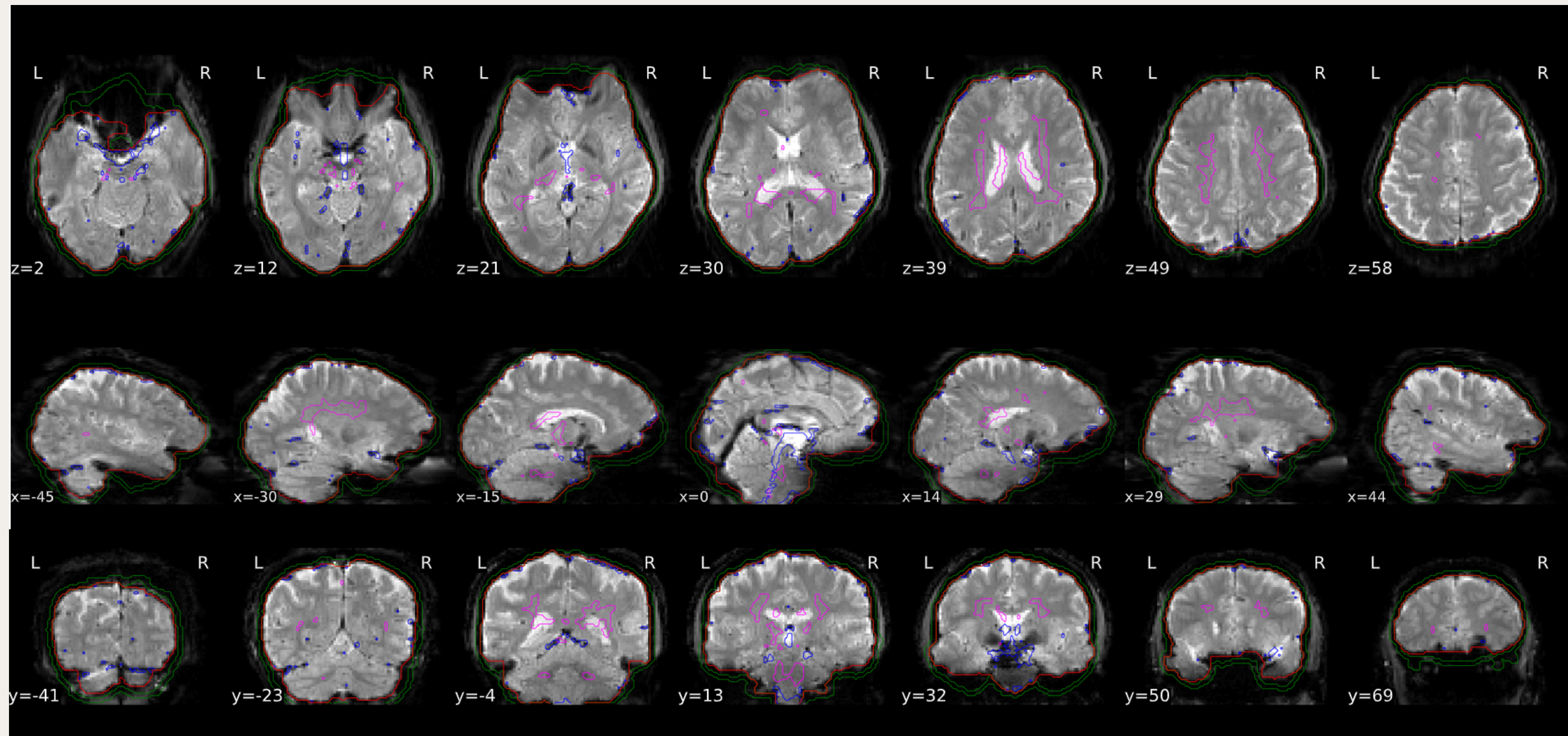
Brain mask and (anatomical/temporal) useful ROIs

The red contour defines the brain region used for analysis.

The magenta region captures non-neuronal signals from white matter & CSF, to model physiological noise (from body)

The blue region selects highly variable voxels, which are likely dominated by noise, for temporal noise correction.

The green region captures edge-related noise such as motion and scanner artifacts.



Observations -

We can't remove noise directly.

Instead, we extract noise signals and add them to our model -

$$\text{BOLD}(t) = X_{\text{audio}}(t)\beta + X_{\text{noise}}(t)\gamma + \epsilon$$

So, then, the model can learn how much of the voxel signal is due to noise and subtracts that part mathematically.

Week - 5

1) Audio feature extraction & Modelling temporal delays -

Aim:-

- Extract audio from tsv files for each run (600sec audio with 40 diff 30sec clips
- Extract audio features
- Create temporal lags for every musical feature

Feature Categories:

- Timbre (MFCCs): 13 Mel-Frequency Cepstral Coefficients to represent the spectral texture of instruments.
- Harmony (Chroma): 12-bin pitch representation (one per semitone) to track melodic and harmonic progression.
- Dynamics (RMS): Root Mean Square energy to quantify perceived loudness.
- Spectral Centroid: A measure of "spectral brightness" (the "center of mass" of the sound frequencies)

```
t = "/content/sub-001_task-Training_run-01_events.tsv"
events = pd.read_csv(t, sep="\t")
print(events)
```

	onset	duration	genre	track	start	end
0	0	15	'disco'	9	4.21	19.21
1	15	15	'blues'	98	6.46	21.46
2	30	15	'hiphop'	53	0.40	15.40
3	45	15	'pop'	29	3.56	18.56
4	60	15	'jazz'	9	7.44	22.44
5	75	15	'classical'	39	7.28	22.28
6	90	15	'disco'	43	1.81	16.81
7	105	15	'country'	54	0.50	15.50
8	120	15	'metal'	92	10.00	25.00
9	135	15	'reggae'	4	4.86	19.86
10	150	15	'rock'	2	8.88	23.88
11	165	15	'blues'	36	7.17	22.17
12	180	15	'jazz'	18	6.69	21.69
13	195	15	'rock'	54	6.09	21.09
14	210	15	'classical'	95	6.46	21.46
15	225	15	'reggae'	68	14.27	29.27
16	240	15	'country'	39	12.05	27.05
17	255	15	'hiphop'	10	8.38	23.38
18	270	15	'pop'	43	10.53	25.53
19	285	15	'metal'	57	13.73	28.73
20	300	15	'disco'	81	4.50	19.50
21	315	15	'blues'	85	12.14	27.14
22	330	15	'rock'	43	8.49	23.49
23	345	15	'classical'	23	5.81	20.81
24	360	15	'disco'	91	14.06	29.06
25	375	15	'hiphop'	66	4.11	19.11
26	390	15	'reggae'	55	13.83	28.83
27	405	15	'pop'	92	11.62	26.62
28	420	15	'metal'	38	10.53	25.53
29	435	15	'jazz'	77	14.24	29.24
30	450	15	'country'	36	14.06	29.06
31	465	15	'jazz'	16	7.88	22.88
32	480	15	'reggae'	26	9.56	24.56
33	495	15	'disco'	66	14.95	29.95
34	510	15	'pop'	17	5.53	20.53
35	525	15	'rock'	37	4.18	19.18
36	540	15	'blues'	77	13.95	28.95
37	555	15	'classical'	84	6.49	21.49
38	570	15	'country'	67	14.77	29.77
39	585	15	'hiphop'	72	12.19	27.19
40	600	15	'metal'	78	1.74	16.74

MFCCs (Mel-Frequency Cepstral Coefficients)

MFCCs approximate how humans perceive sound using Mel scale

mel $\rightarrow$ STFT with triangular window

cochleo $\rightarrow$ STFT with gaussian wiindow

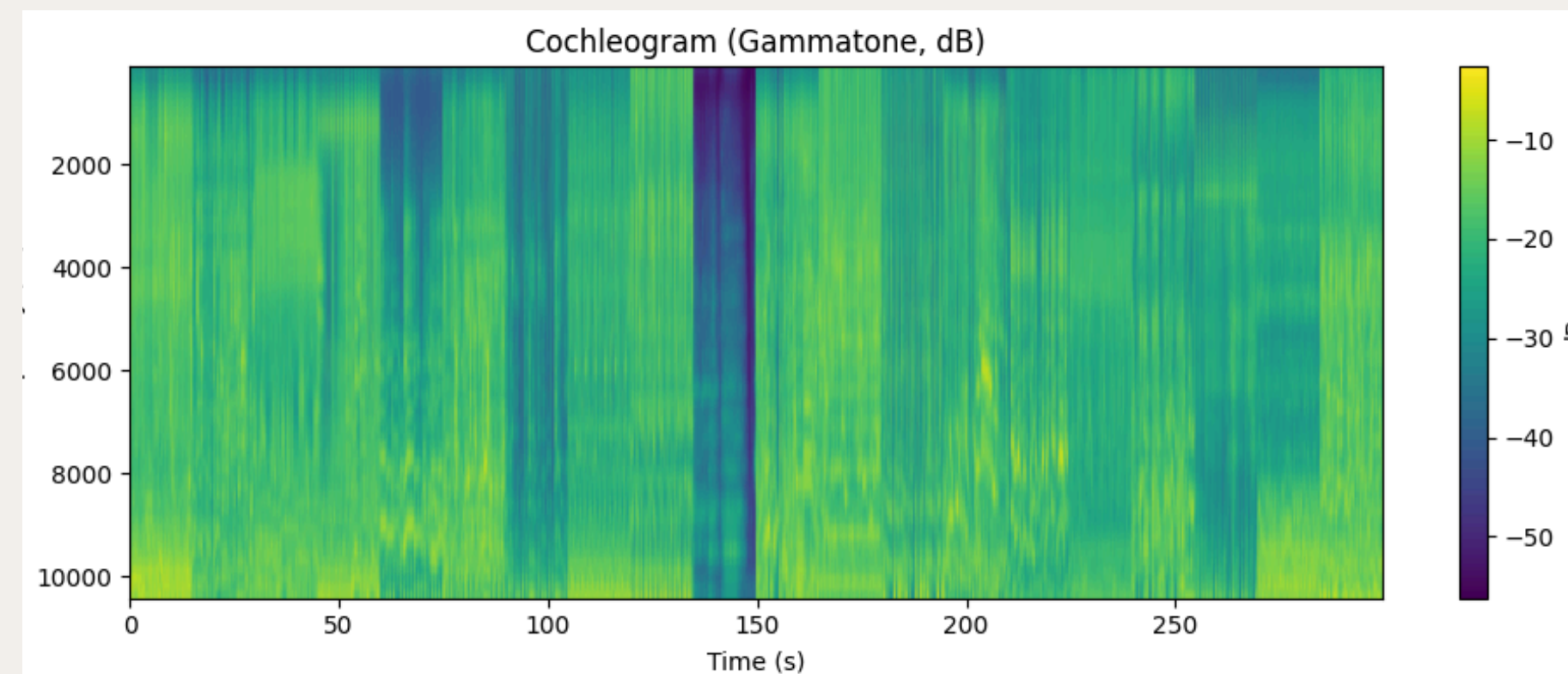
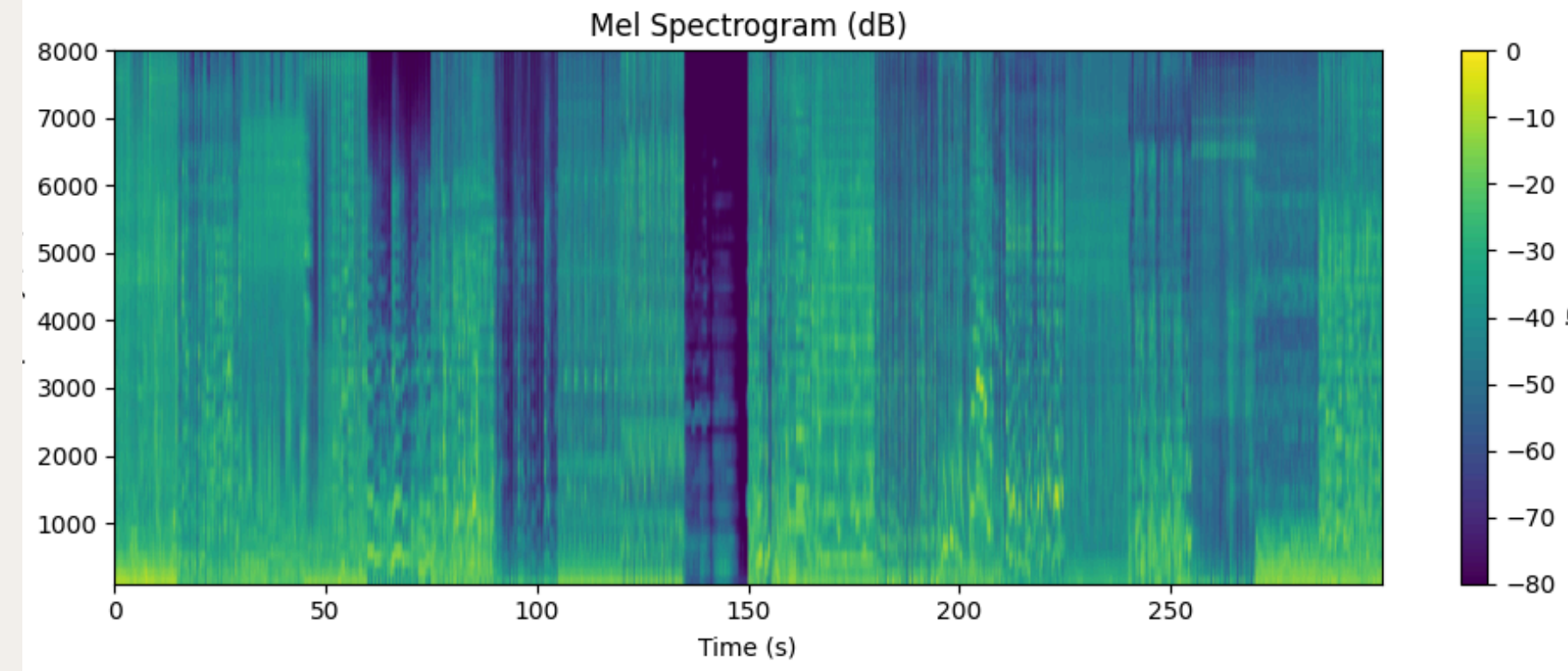
Cochleogram has Biological realism but for fast computation we are going with mel-spectrogram

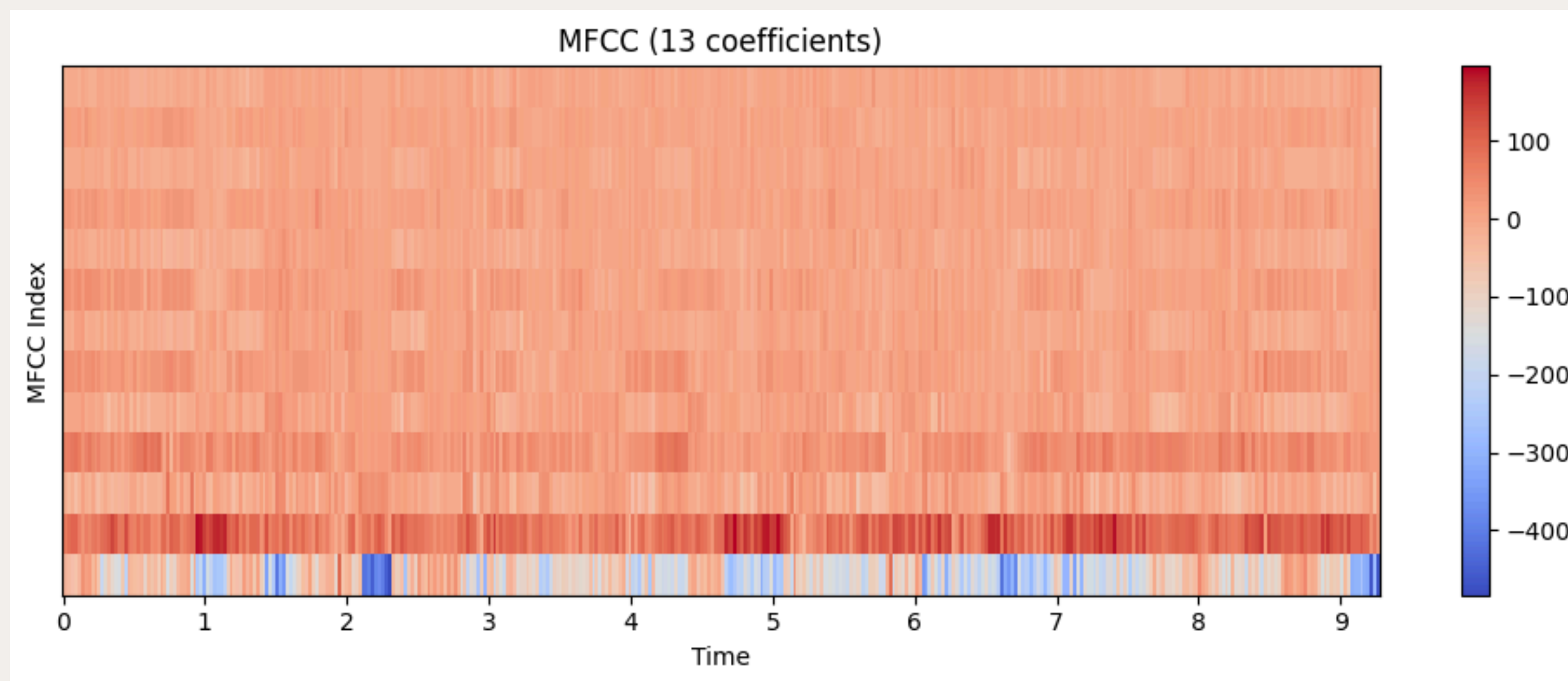
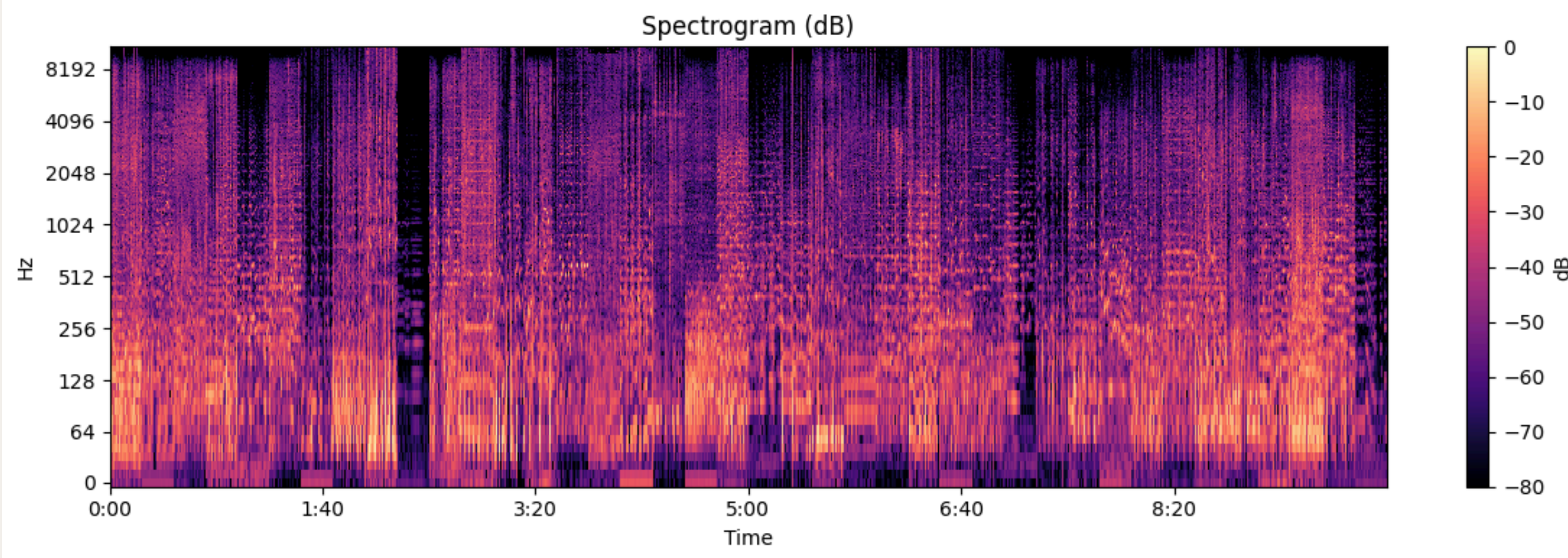
mel scale:

$$m = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$

Cepstrum:

$$C_r = \mathcal{F}^{-1} \{ \log(|\mathcal{F}\{f(t)\}|) \}$$





Chroma Features

Instead of full spectrum → map frequencies into 12 bins [C, C#, D, D#, E, F, F#, G, G#, A, A#, B]

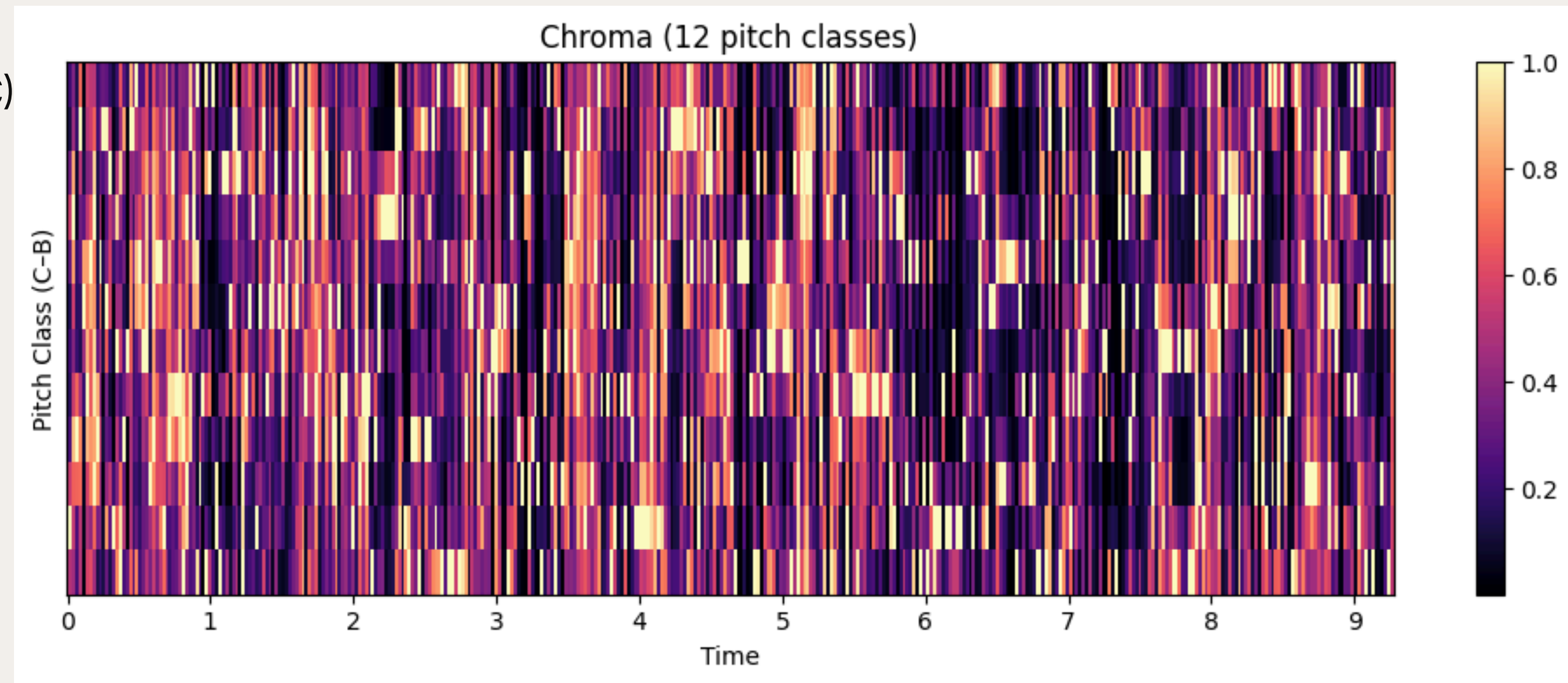
Pitch class distribution → captures Harmony

All frequencies that are octave multiples collapse into one bin.

Example:

- 261 Hz (C4)
- 523 Hz (C5)
- → both go into same chroma bin (C)

$$\text{chroma} = \log_2(f/f_{ref}) \mod 1$$



RMS Energy

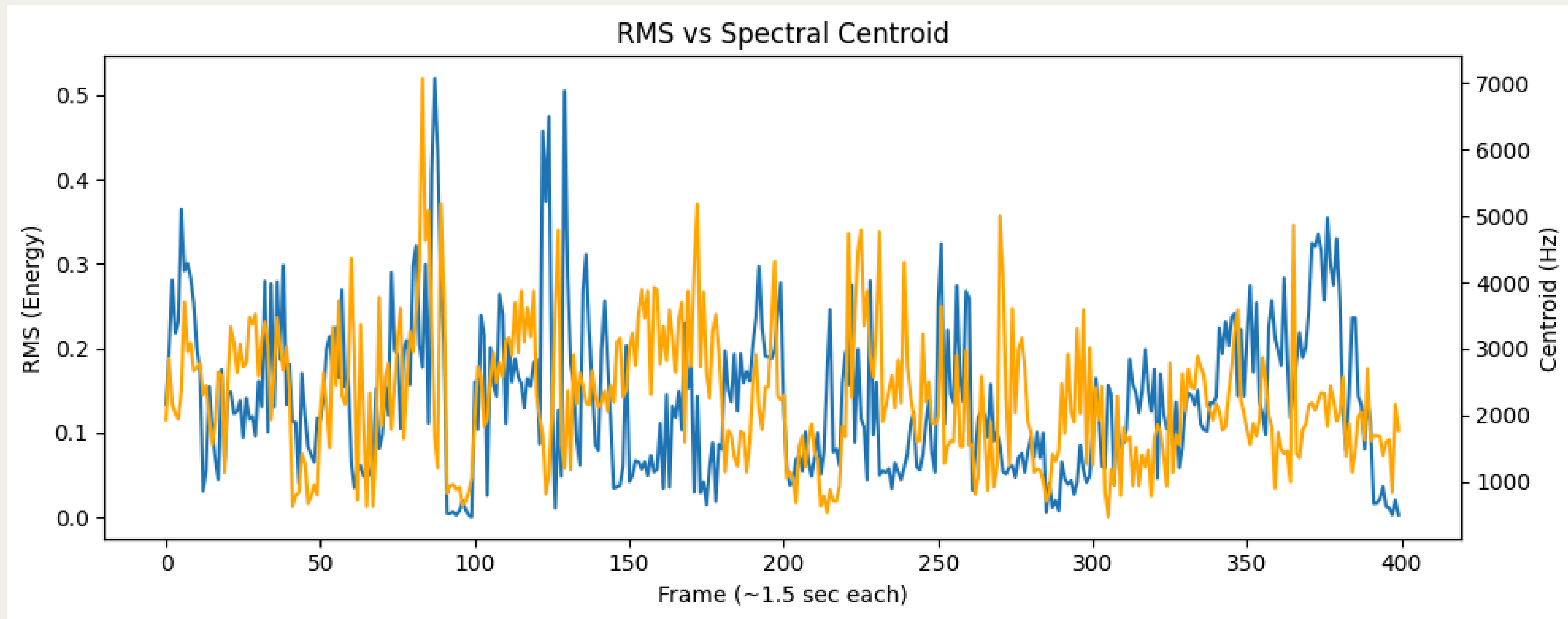
capture → Perceived loudness / signal power

$$RMS = \sqrt{\frac{1}{N} \sum_{n=1}^N x[n]^2}$$

Spectral Centroid

capture → Where the energy is concentrated in frequency

$$C = \frac{\sum f_k \cdot S(f_k)}{\sum S(f_k)}$$



Neural activity is instantaneous, but the BOLD (Blood-Oxygen) signal peaks with a 4–6 second latency.

- 1) Instead of assuming one fixed delay, we created 6 temporal lags for every musical feature
- 2) Multiplied the 27 base features by 6 delays [1.5, 3, 4.5, 6, 7.5, 9] sec, resulting in a 162-dimensional feature space.
- 3) Hence, we use a multi-lag model to capture the full echo of the hemodynamic response instead of just one specific moment.

Shape: (400, 162)

	MFCC_0_t-1.5s	MFCC_1_t-1.5s	MFCC_2_t-1.5s	MFCC_3_t-1.5s	MFCC_4_t-1.5s	MFCC_5_t-1.5s	MFCC_6_t-1.5s	MFCC_7_t-1.5s	MFCC_8_t-1.5s	MFCC_9_t-1.5s	...	Chroma_E_t-9s	Chroma_F_t-9s	Chroma_F#_t-9s	Chroma_G_t-9s	Chroma_G#_t-9s	Chroma_A_t-9s	Chroma_A#_t-9s	Chroma_B_t-9s
0	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	...	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
1	-89.454430	103.433762	-14.477322	53.373802	0.727541	31.391712	3.185943	23.764351	-1.761361	13.154818	...	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
2	-27.653975	70.573547	-18.027283	60.491665	-5.165735	35.227261	-14.188894	37.336555	-12.542707	19.758047	...	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
3	-42.547909	89.433182	-34.071579	64.119568	0.735227	31.688515	-7.069813	31.952663	-0.662291	26.062618	...	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
4	-54.951603	101.964325	-17.758818	73.930687	-2.091931	34.132160	-12.576090	18.488842	-8.278286	16.034754	...	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
395	-255.980804	119.404671	-16.800768	40.608746	13.178621	4.413335	-8.821210	-10.959431	-2.531289	5.998221	...	0.357331	0.406116	0.710328	0.241449	0.128731	0.213098	0.461027	1.000000
396	-308.383759	102.151260	-1.470080	37.247223	15.096527	4.857159	-2.970495	1.888837	6.275039	4.737465	...	0.219910	0.312207	1.000000	0.294596	0.065035	0.162938	0.394651	0.740544
397	-319.277924	96.274414	2.201954	30.999321	6.076867	4.135155	-5.722764	-1.648193	6.509119	4.648037	...	0.161211	0.106192	0.251933	0.089986	0.086832	0.460762	0.197659	0.145364
398	-463.834290	33.983963	28.559772	23.193125	19.571962	17.234880	15.107635	12.708254	10.053970	7.442232	...	0.113394	0.088684	0.641165	1.000000	0.102029	0.048476	0.130201	0.077777
399	-283.986816	75.407120	-42.898621	17.246525	-1.607304	5.292273	1.107313	24.368334	15.636175	8.774566	...	0.202325	0.087619	0.126023	0.182045	0.082117	0.127443	0.150433	0.256454

400 rows × 162 columns

2) Data Dimensionality Reduction and Confound regression -

Data dimensionality reduction -

- 1) Input (4D): A temporal sequence of 3D volumes ($X \times Y \times Z \times \text{Time}$)
- 2) The Filter (3D Mask): A spatial lookup table that identifies Gray Matter coordinates in MNI152 space.
- 3) The Output (2D Array): A flattened Voxel-by-Time matrix.

Confound regression -

- 1) Utilized fMRIPrep's metadata to identify 9 key noise sources (6 motion axes + Global, CSF, and White Matter signals).
- 2) Performed Linear Regression to mathematically subtract these "noise signatures" from every voxel's activity.

3) Voxel-wise encoding and model optimization -

1) Ridge Regression : A stable linear model that handles the high-dimensional music data while ignoring random scanner noise.

The model is trained for each of the 238,912 voxels independently to create a high-resolution map of the auditory cortex.

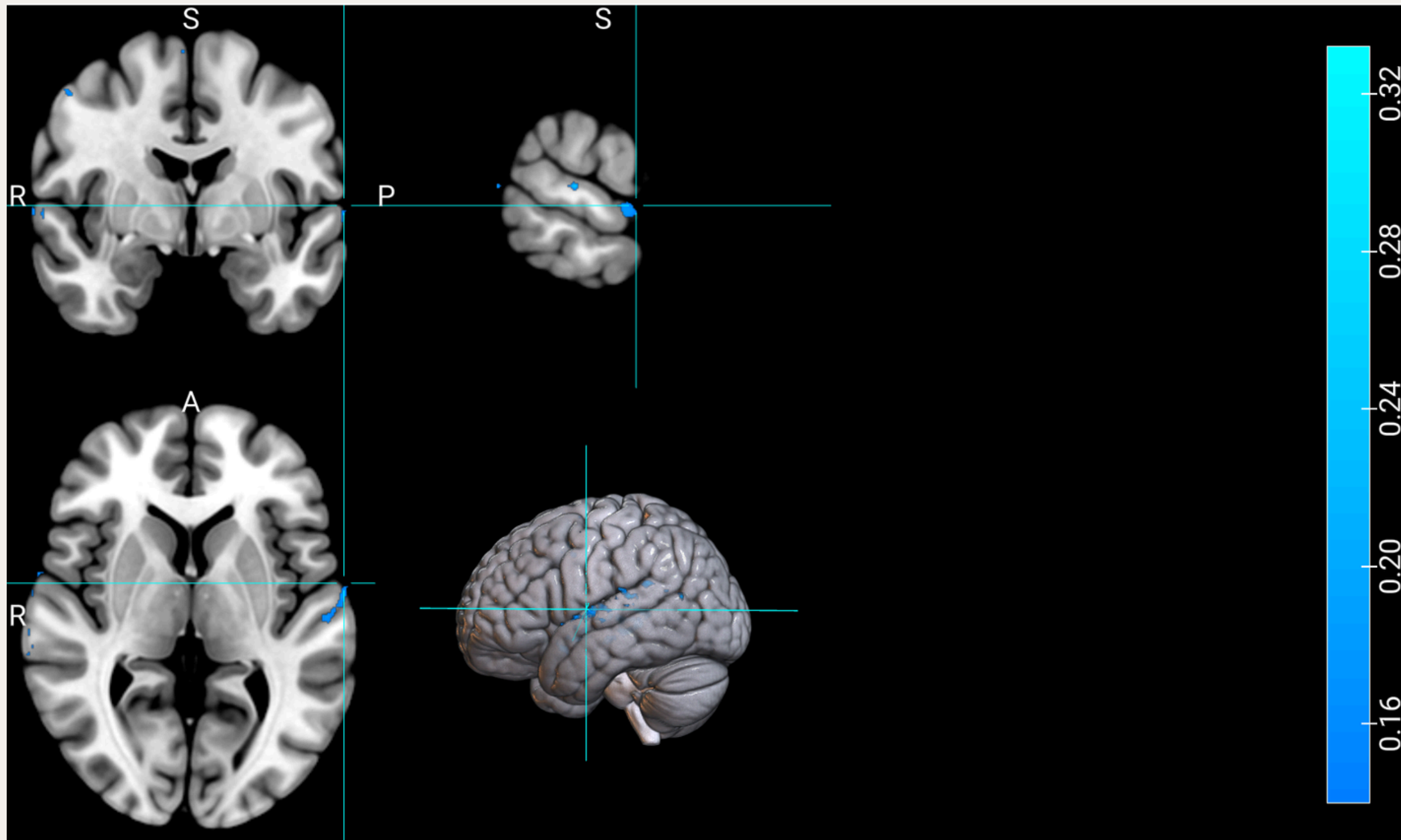
2) We measure Pearson Correlation (r) values. They measure how well the model's "predicted brain activity" (using test runs) matches the "real brain activity" (using training runs) from the scanner.

Maximum correlation we get $r = 0.33$: The brain activity is significantly similar enough to prove the music is driving the brain, but they aren't identical because of the "Noise" (the brain's other thoughts).

3) High-accuracy voxels were centered in the Heschl's Gyrus and the Superior Temporal Gyrus (STG), validating the model's precision.

Brain mask and brain tissue segmentation of the T1w

The model successfully 'decoded' the subject's brain, explaining up to 32% of the activity in the auditory cortex using only music features. 3D mapping reveals that the auditory hotspots are anatomically precise, proving that Librosa features are a powerful proxy for human music perception.



Mapping acoustic features to 3 categories -

1) TIMBRE -

- Timbre refers to the texture of the sound - or what is making it i.e the instrument, source.
- Acoustic feature - **MFCCs**
- Brain region - **Primary auditory cortex/Hesch's gyrus**

2) HARMONY -

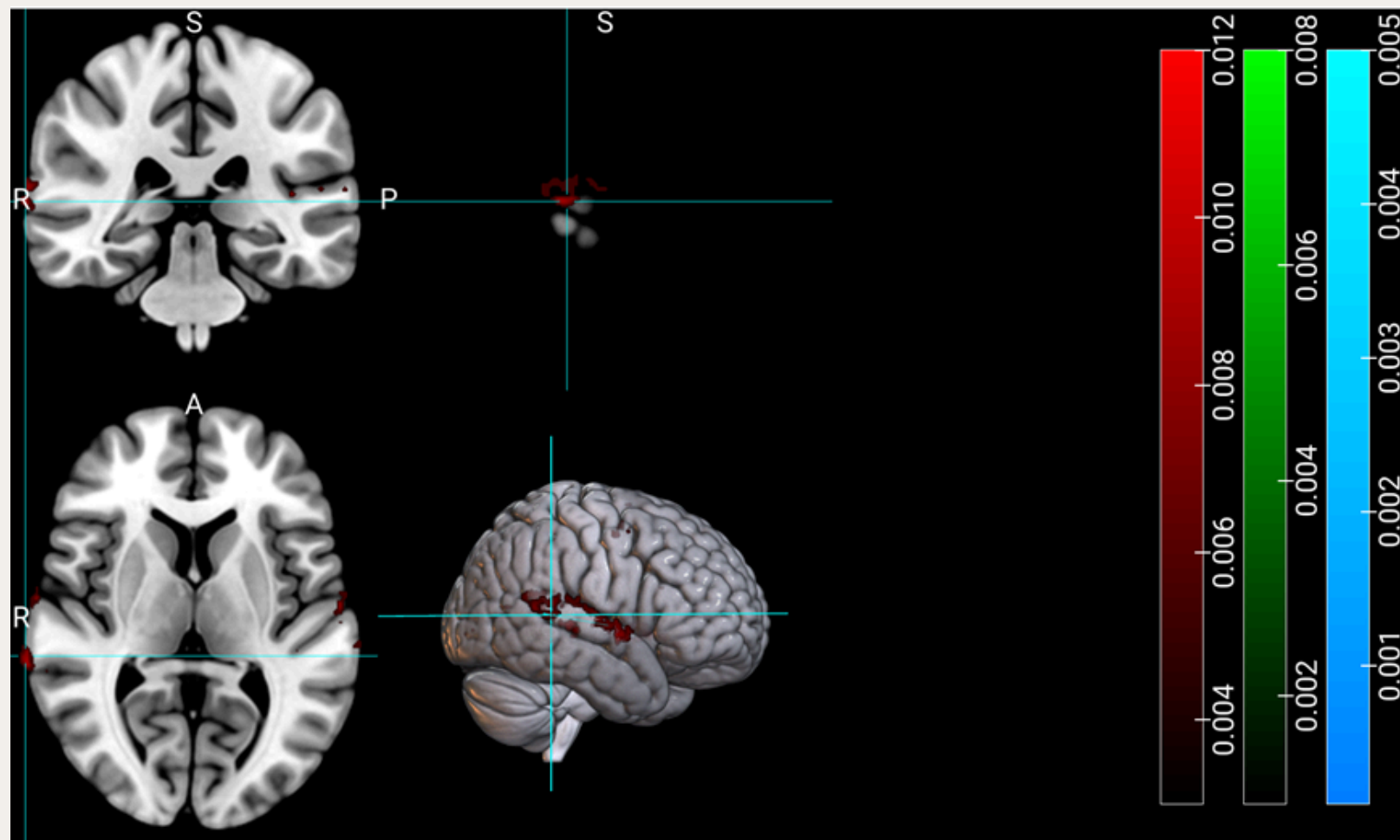
- Harmony refers to the exact notes and chords in the music.
- Acoustic feature - **Chroma (Mapping the 12-note scale)**
- Brain region - **Superior Temporal Gyrus**

3) RHYTHM -

- Rhythm refers to the timing, beat, and energy of the sound.
- Acoustic Feature: **RMS Energy & Spectral Flux** (Volume peaks and sudden changes).
- Brain Region: **Posterior STG**

Voxel wise sensitivity to acoustic features

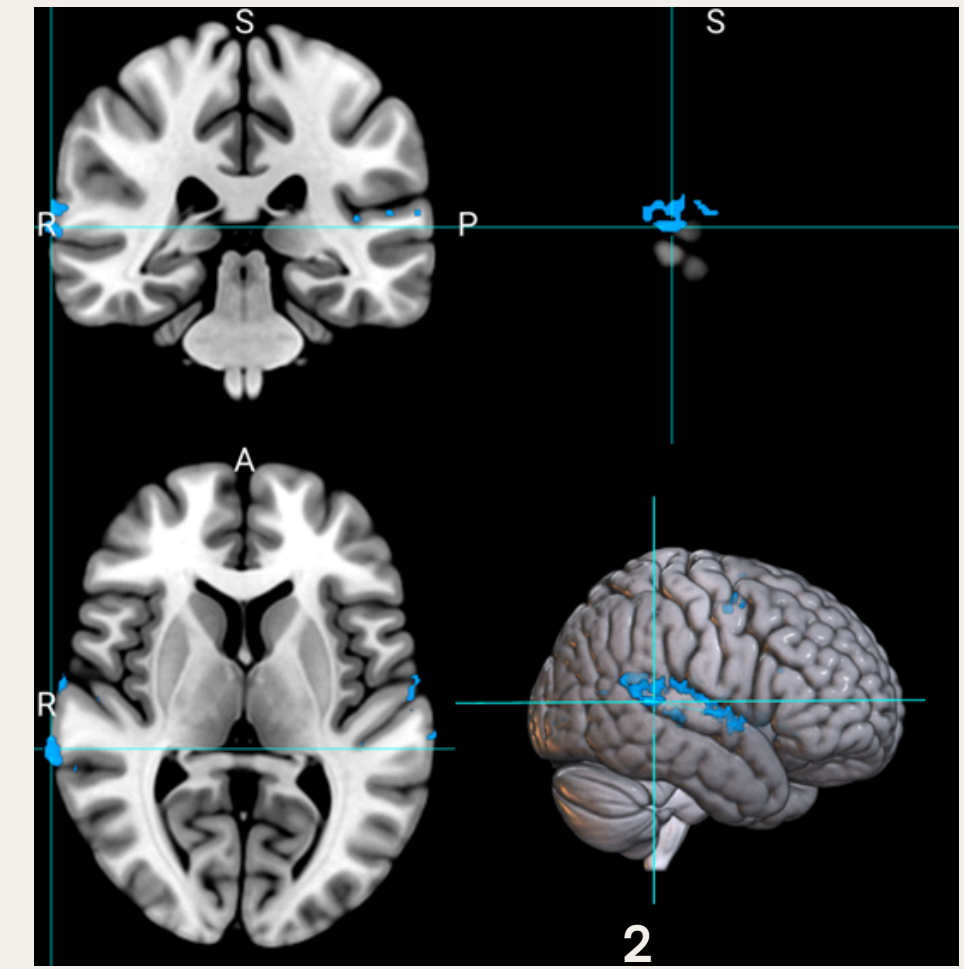
- These maps were generated by extracting the mean absolute beta weights from our Ridge Regression model. We then applied a significance mask based on our prediction accuracy ($r > 0.15$) to ensure we were only visualizing voxels with a statistically reliable stimulus-response relationship.
- While the individual sensitivity maps show broad spatial overlap across the auditory ROI, indicating that these regions are multi-sensory, the magnitude of the weights differs significantly. Harmony features consistently exhibit the highest encoding strength.



Harmony



Rhythm

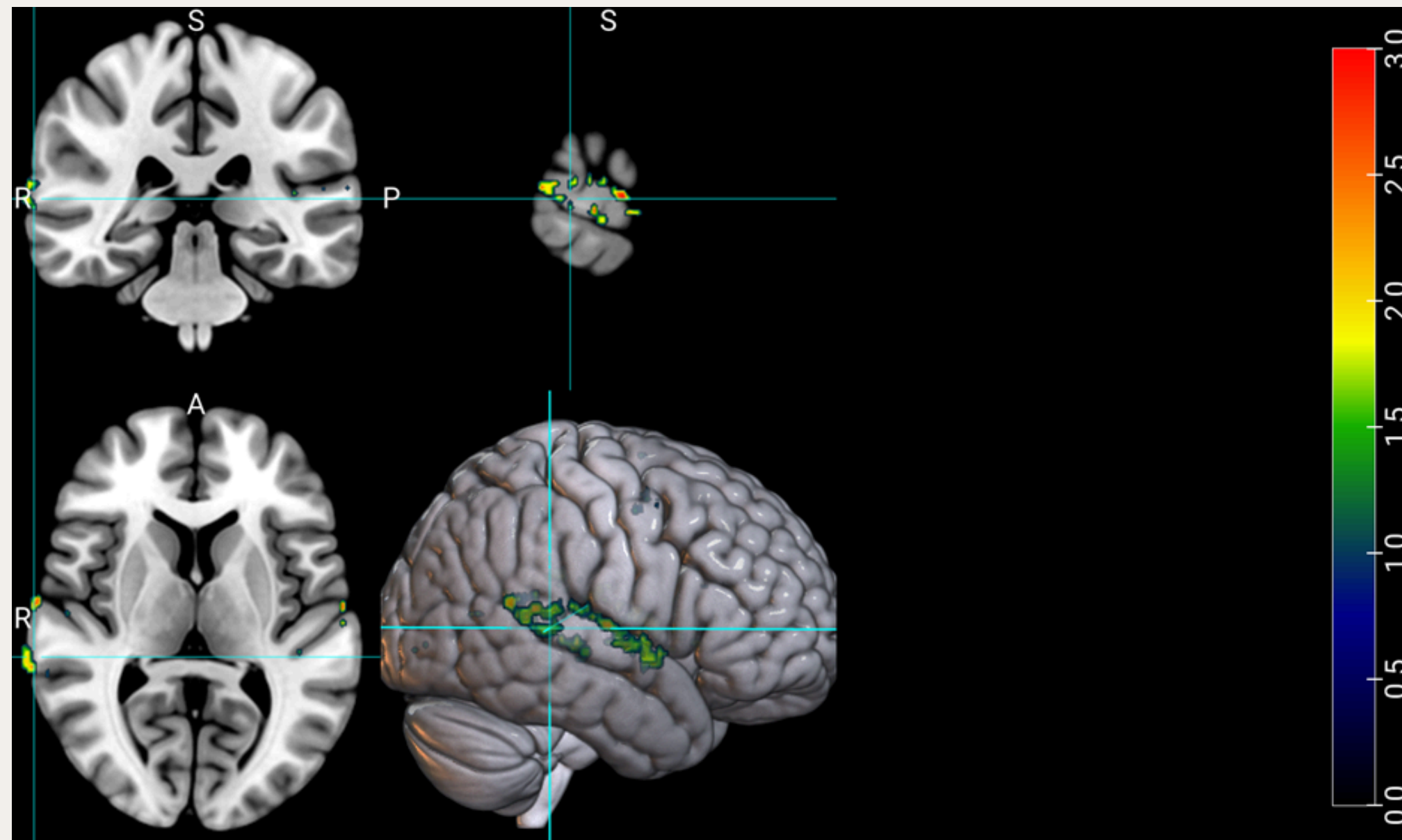


Timbre

Mapping the Neural Territories of Timbre, Harmony, and Rhythm

While these musical dimensions show overlapping neural populations in the Superior Temporal Gyrus, Winner-Take-All analysis reveals that Harmony is the primary driver of neural activity in this subject, occupying the largest functional territory, followed by specialized clusters for Timbre in the core and Rhythm in the posterior regions.

- The Blue (Timbre) sits lower/deeper.
- The Green (Harmony) spreads across the middle belt.
- The Red (Rhythm) sits in specific clusters, often toward the back.



Genre-wise analysis

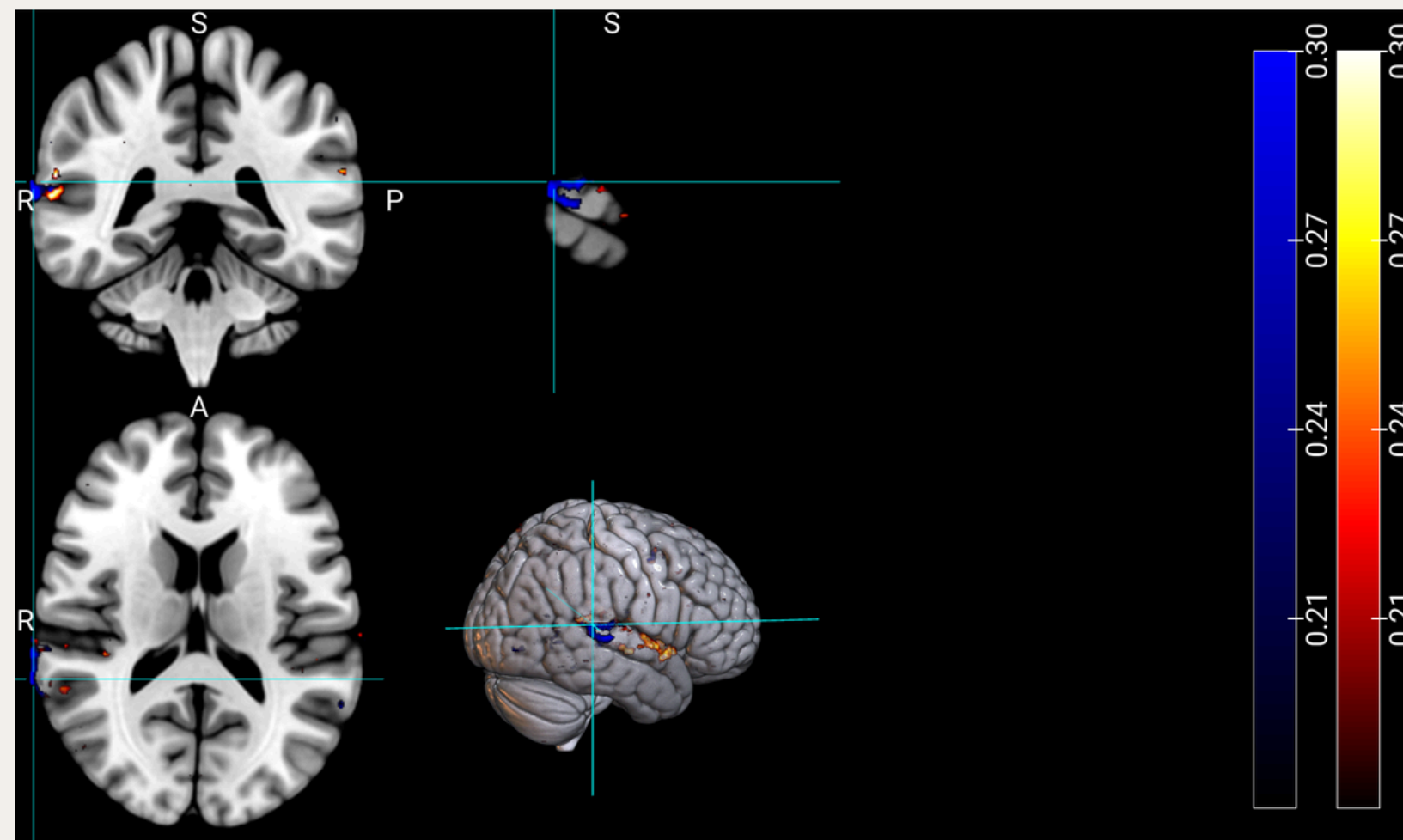
We did genre wise analysis using accuracy maps -

- Universal Training: A "Master Model" is trained on diverse music runs to learn the general relationship between acoustic features and brain activity.
- Genre Isolation: The test data is segmented using timestamps to isolate brain activity recorded specifically during each genre (e.g., Pop vs. Metal).
- Blind Prediction: The model uses the musical features of a specific genre to "guess" how the brain should react based on its prior training.
- Accuracy Mapping (r): Predictions are compared to actual fMRI data; voxels where the guess matches reality ($r > 0.18$) are mapped as "Accuracy Hotspots."

Metal v/s Pop

Your brain doesn't treat all music equally. It "retunes" its auditory networks to focus on the most informative elements of a genre.

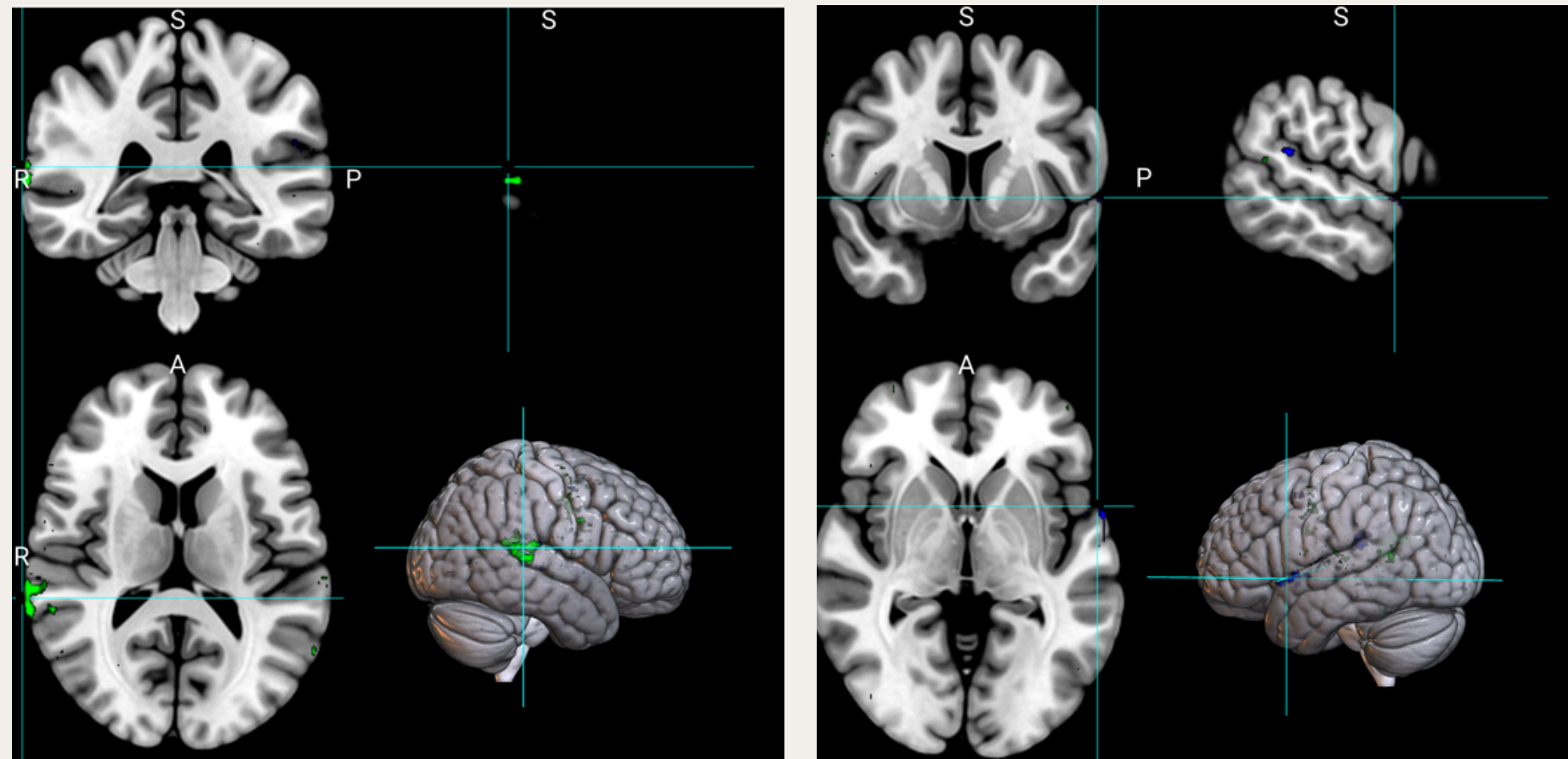
- Metal (Blue): This activity is concentrated in the Posterior STG (toward the back). This area is a specialist in Acoustic Complexity. Metal has high-density textures and rapid transients that require this "back-office" processing power.
- Pop (Warm): This activity moves toward the Anterior/Lateral STG (toward the front/side). This area is often associated with Melodic Expectations and vocal-like patterns. Pop is designed to be catchy and follows melodic "rules" that these front areas are tuned to track.



Classical v/s Disco

Your brain doesn't treat all music equally. It "retunes" its aud

- Disco (Green):
 - Localization: Right Hemisphere - Posterior STG
 - Disco is defined by its "vibe" - the blend of synthesizers, bass, and vocals. The right hemisphere specializes in Timbre (sound quality), allowing the brain to track the rich, layered "color" of the disco dance floor as a single, cohesive groove.
- Classical (Blue): The "Analytical Architect"
 - Localization: Left Hemisphere - Anterior STG
 - Classical music follows mathematical "rules" of harmony and counterpoint. The left hemisphere processes these sequences similarly to language, "decoding" the complex, fast-changing melodic instructions as a structured logical stream.



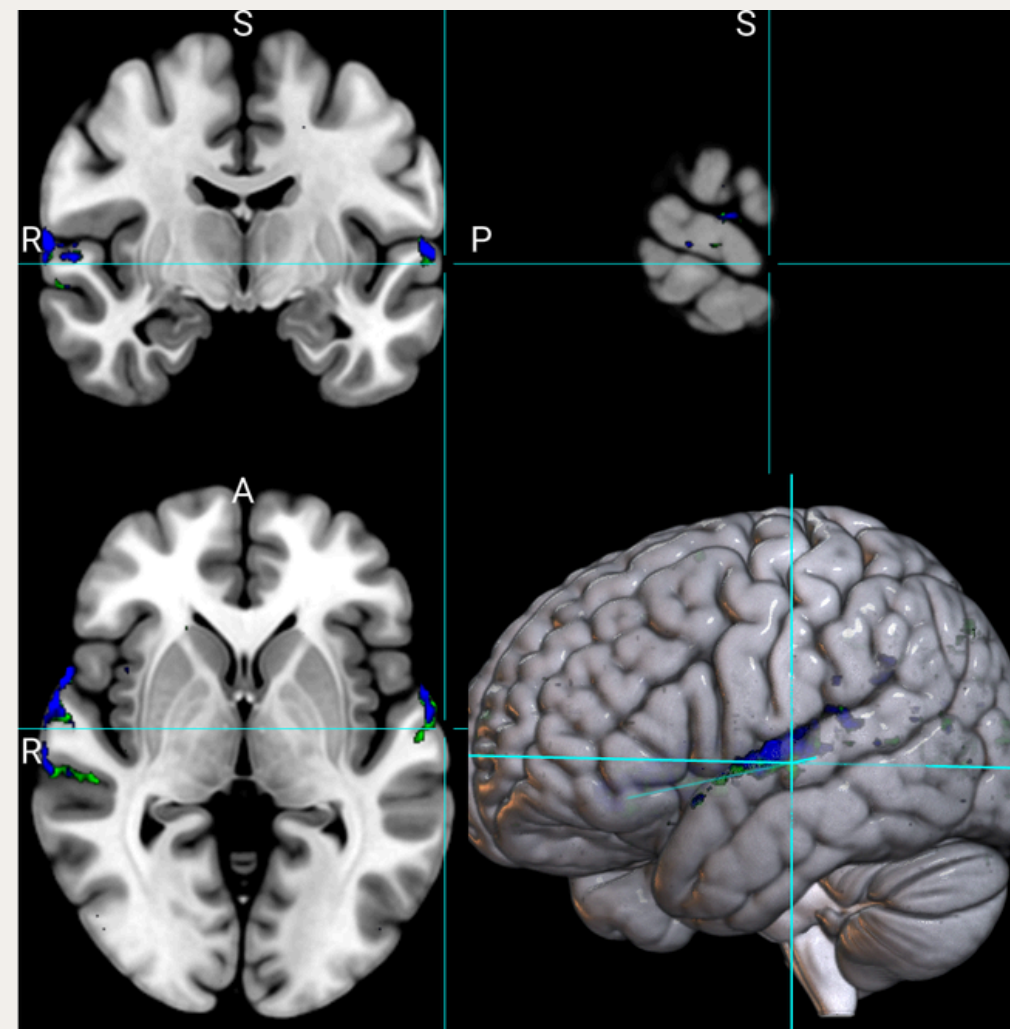
Hip-hop v/s reggae

Hip-Hop (Green) -

- Localization: Tucked deeper into the folds within the Sylvian fissure (Heschl's gyrus)
- Hip-Hop is defined by sharp, raw bursts of sound.
- This suggests the encoding model is primarily tracking Low-Level Acoustic Transients-the raw physical impact of the beat-which is handled by the Primary Auditory Core.

Reggae(Blue) -

- Localization: Outer Folds (the lateral surface of the Superior Temporal Gyrus)
- Reggae is "laid-back." It focuses on Spectral Content. The outer surface is where the brain does its most complex "thinking" about music. Reggae's syncopated, melodic structure requires this higher-level processing power.



Conclusion

- Our model demonstrates that the brain doesn't just "hear" music; it physically retunes its auditory networks to match a genre's most informative features.
- By mapping these activations, we show a clear hierarchy: percussive, high-energy genres like Hip-Hop and Metal anchor deep in the Primary Core to track raw physical timing, while structurally complex genres like Classical and Reggae move to the Outer Ridges to integrate melody and groove.
- This spatial and hemispheric "division of labor" - splitting tasks between a Left-brain "Clock" for rhythm and a Right-brain "Palette" for texture - proves that the auditory cortex is a dynamic filter that reconfigures itself in real-time to decode the specific "logic" of every genre.